The complete string C-group counts for S_8

Human Compression project

We recover the S_8 row of Table 1 in Leemans' survey [1], using the independently reviewed finite enumeration method [2]. The survey identifies generating sets under group automorphisms. Here an ordered representation retains the order of its marked involutions; duality reverses that order. We verify below why these conventions give the stated comparison.

Theorem 1. *The numbers of string C-group representations of S_8 are as follows.*

rank	3	4	5	6	7
ordered, modulo automorphisms	126	71	20	2	1
self-dual ordered classes	10	1	2	0	1
modulo automorphisms and duality	68	36	11	1	1

There are no representations of larger rank.

Proof. Every nonidentity involution in the natural action is a product of e disjoint transpositions for exactly one $e \in \{1, 2, 3, 4\}$. The four possibilities are the four initial branches of the acquired enumeration. Each branch begins with the specified representative, and the exact centralizer acts on all possible subsequent marks. The complete method proves that its successive orbit representatives exhaust the ordered tuples up to simultaneous point conjugacy. It also pays every commutation restriction, the actual subgroup intersection test and the pruning of prefixes which cannot generate S_8. The inherited independent-generator bound gives rank at most seven; the four branches are therefore complete for this group. All four terminate, without reaching their time limits, with the first row above.

We independently check each of the 220 retained tuples. Its marks are nonidentity involutions with the required distant commutations, and their generated subgroup has order 40320, hence is S_8. For every pair of subsets I, J of its marks, we explicitly form the two generated subgroups and verify

$$\langle \rho_i : i \in I \rangle \cap \langle \rho_j : j \in J \rangle = \langle \rho_k : k \in I \cap J \rangle.$$

This performs 71 296 actual subgroup-intersection checks. These checks validate the returned objects; completeness is supplied by the four finished enumerations and the acquired method, not by the number of positive objects alone.

For clarity we also verify the class identifications outside the group enumerator. Root the coloured permutation graph at a point, discover vertices in breadth-first order, and at each vertex inspect the marks in their given order. Write every marked permutation in this new numbering and take the least resulting array over the eight roots. This array is invariant under point conjugacy. Conversely equality of two arrays supplies the required point bijection, so it is a complete invariant. This is the unrooted version of the previously acquired coloured traversal [3]. The 220 arrays are distinct within their ranks. Reversing each tuple matches exactly one retained array, and the induced permutation of the classes is an involution with the fixed-point counts in the second row. Thus its orbit counts are $(126 + 10)/2$, $(71 + 1)/2$, $(20 + 2)/2$, $(2 + 0)/2$ and $(1 + 1)/2$, respectively. Finally, every automorphism of S_8 is inner. The exact finite check also verifies $|\operatorname{Aut}(S_8)| = 40320$; since $Z(S_8) = 1$, point conjugacy already accounts for all automorphisms. Adjacent marks cannot commute, by the acquired direct-factor argument for S_8. Thus the graph of noncommuting marks is a path. An isomorphism of their generating sets preserves this path, whose only automorphisms are the identity and reversal. Consequently the last row is exactly the source's isomorphism convention. □

Evidence and attribution

The counts are existing results reported in the source, not new classifications. The acquired common method contains the actual enumeration proof and its small controls. This component executes that unchanged method for the four fixed S_8 branches and supplies the complete literal output, independent

subgroup validation and reversal maps. It does not import an unverified source count as a premise. All arithmetic and subgroup operations use exact permutations in GAP; the additional canonicalization uses only Python integer arrays. The separate checker reruns the four branches, reconstructs their literal inputs, and repeats both validations. A timeout, damaged receipt or missing terminal marker gives no certificate.

The ranks six and seven were already covered by the acquired stable counts. The present result newly pays the three source cells in ranks three, four and five. The rest of the survey contract, including the larger unrecovered experimental rows, remains separate. No whole-source admission, compression measurement or new conjecture resolution follows from this bounded component.

References

[1] D. Leemans, *String C-group representations of almost simple groups: a survey*, fixed arXiv:1910.08843v2, Table 1, physical page 6.

[2] Human Compression project, *Complete permutation searches and the six-point row*, accepted finite-method component, with its distinct review and root adoption; exact frozen inputs accompany this note.

[3] Human Compression project, accepted balanced-pair component for the high-rank counting theorem, literal canonicalization argument and its retained independent review.